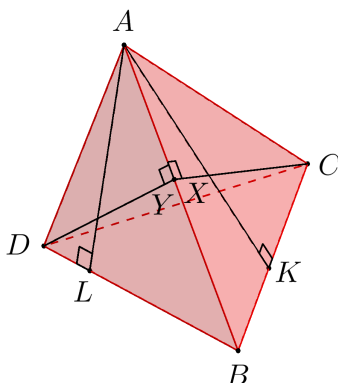


Problem 33 (Oral Moscow Team MO 2016 Grade 11 Problem 3). All faces of the tetrahedron $ABCD$ are acute-angled triangles. The altitudes AK and AL of $\triangle ABC$ and $\triangle ABD$ are drawn where K and L are the feet of altitudes. It is known that the points C, K, L and D lie on the same circle. Prove that AB is perpendicular to CD .

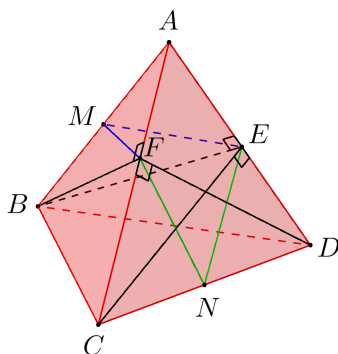


Solution. Let CX and DY be the altitudes of $\triangle ABC$ and $\triangle ABD$ where X and Y are the feet of altitudes. Since the points in each of the groups $\{C, K, L, D\}$, $\{A, X, K, C\}$ and $\{A, Y, L, D\}$ are concyclic, we have

$$BX \cdot BA = BK \cdot BC = BL \cdot BD = BY \cdot BA.$$

As X and Y lie on the segment AB , this implies $X = Y$. Thus, AB is perpendicular to both CX and DX , meaning that it is perpendicular to the plane CXD . Hence, $AB \perp CD$. $\square$

Problem 34 (Romania District Olympiad 2013 Grade 8 Problem 4). Let $ABCD$ be a tetrahedron such that $AD \perp BC$ and $AC \perp BD$. Let E and F be the projections of point B on the lines AD and AC , respectively. Let M and N be the midpoints of AB and CD , respectively. Show that $MN \perp EF$.



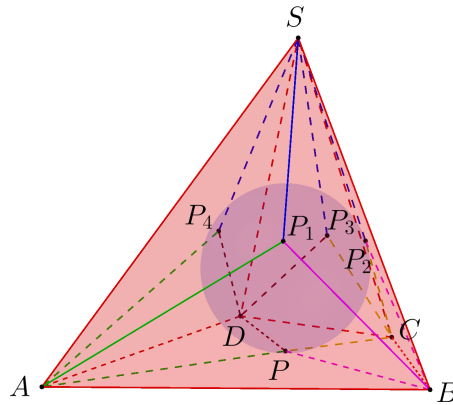
Solution. We show that both M and N lie on the mediator of EF .

Firstly, consider the right-angled $\triangle ABF$. Since M is the midpoint of the hypotenuse AB , we have $MF = \frac{1}{2}AB$. By symmetry, $ME = \frac{1}{2}AB = MF$. So M lies on the mediator of EF .

Secondly, since $AC \perp BD$ and $AC \perp BF$, the line AC is perpendicular to the plane BDF . Thus, $AC \perp DF$. Again, since N is the midpoint of the hypotenuse CD of the right-angled $\triangle CDF$, we have $NF = \frac{1}{2}CD$. By symmetry, $NE = \frac{1}{2}CD = NF$. So N lies on the mediator of EF .

It is now obvious that $MN \perp EF$. $\square$

Problem 35 (Ukraine Champions Tournament 2008 Senior Problem 4). Let $SABCD$ be a quadrangular pyramid whose base $ABCD$ is a convex quadrilateral. Suppose there is an inscribed sphere in the pyramid, and let P be the contact point of this sphere with the base $ABCD$. Prove that $\angle APB + \angle CPD = 180^\circ$.

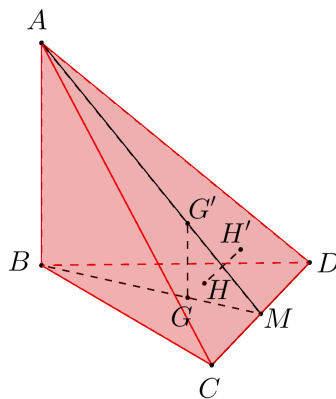


Solution. Let the insphere touch the faces SAB, SBC, SCD, SDA at P_1, P_2, P_3, P_4 respectively. Then $\triangle PAB \cong \triangle P_1AB$, $\triangle PBC \cong \triangle P_2BC$, $\triangle PCD \cong \triangle P_3CD$, $\triangle PDA \cong \triangle P_4DA$, $\triangle P_1SA \cong \triangle P_4SA$, $\triangle P_2SB \cong \triangle P_1SB$, $\triangle P_3SC \cong \triangle P_2SC$ and $\triangle P_4SD \cong \triangle P_3SD$. It follows that

$$\begin{aligned}
 & (\angle APB + \angle CPD) - (\angle BPC + \angle DPA) \\
 &= \angle AP_1B + \angle CP_3D - \angle BP_2C - \angle DP_4A \\
 &= (\angle AP_1B + \angle AP_1S + \angle BP_1S) + (\angle CP_3D + \angle CP_3S + \angle DP_3S) - (\angle BP_2C + \angle BP_2S + \angle CP_2S) \\
 &\quad - (\angle DP_4A + \angle DP_4S + \angle AP_4S) \\
 &= 180^\circ + 180^\circ - 180^\circ - 180^\circ = 0^\circ.
 \end{aligned}$$

Also, it is clear that $(\angle APB + \angle CPD) + (\angle BPC + \angle DPA) = 360^\circ$. So $\angle APB + \angle CPD = \angle BPC + \angle DPA = 180^\circ$. $\square$

Problem 36 (Romanian MO 2012 Grade 8 Problem 3). In a tetrahedron $ABCD$, $\triangle ACD$ and $\triangle BCD$ are acute-angled triangles. Let G and H be the centroid and the orthocentre of $\triangle BCD$. Let G' and H' be the centroid and the orthocentre of $\triangle ACD$. Given that HH' is perpendicular to the plane ACD , show that GG' is perpendicular to the plane BCD .



Solution. We first show that AB is perpendicular to the plane BCD .

As $AC \perp HH'$ and $AC \perp DH'$, AC is perpendicular to the plane DHH' . This shows $AC \perp DH$. Together with $DH \perp BC$, we know that DH is perpendicular to the plane ABC . Thus, $AB \perp DH$. Similarly, $AB \perp CH$. This means AB is perpendicular to the plane CDH as claimed.

Now, let M be the midpoint of CD . Then AG' and BG meet at M , and $\frac{AG'}{G'M} = \frac{BG}{GM} = 2$. Thus, $\triangle MG'G \sim \triangle MAB$, and so GG' is parallel to AB . Thus, GG' is perpendicular to the plane BCD . $\square$